

Cellular Expert Geodata Requirements

2025

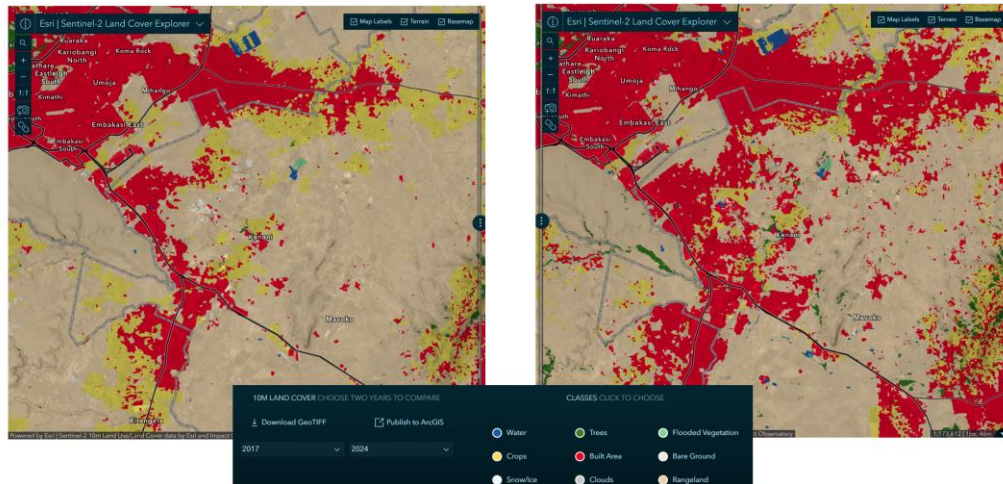


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1. High-Precision Inputs and Next-Generation Propagation

CE Express is designed to work with any geospatial data available to the customer and fully exploit its precision for the most accurate coverage and QoS calculations. The platform supports multi-resolution input datasets — from freely available global sources such as Sentinel-2 10 m land cover and ASTER DEM, to premium high-resolution terrain and 3D building models when provided by the customer or government agencies.



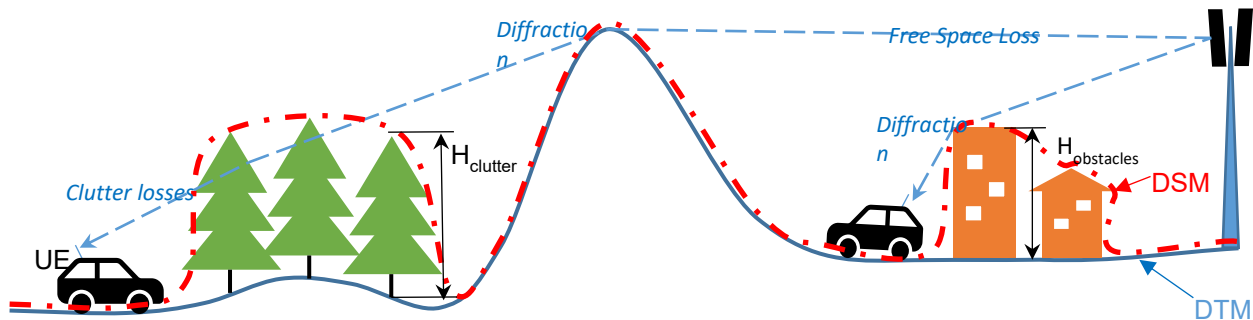
Source: <https://livingatlas.arcgis.com/landcoverexplorer/>

By leveraging whatever data is available locally, CE Express performs nationwide calculations at the maximum feasible resolution, accurately modeling signal propagation even in dense urban environments. Support for 3D multi-height calculations ensures that coverage predictions reflect street-level, indoor, and rooftop conditions, providing regulators with a realistic representation of service availability.

This flexibility ensures that customers can use their existing GIS assets, open datasets, or commercial data they already license, turning them into actionable broadband maps without additional data procurement requirements from the solution provider.

By using terrain elevation, obstacles, and clutter classification in every calculation, Cellular Expert accurately models:

- **Line-of-Sight and Non-Line-of-Sight Conditions** – Determining diffraction, reflection, and shadowing effects over hills, valleys, and urban obstacles.
- **Coverage Footprints** – Generating precise signal strength maps at national, regional, and local levels.
- **Capacity and Interference Analysis** – Modeling realistic signal overlaps and interference zones for multi-operator, multi-technology environments.

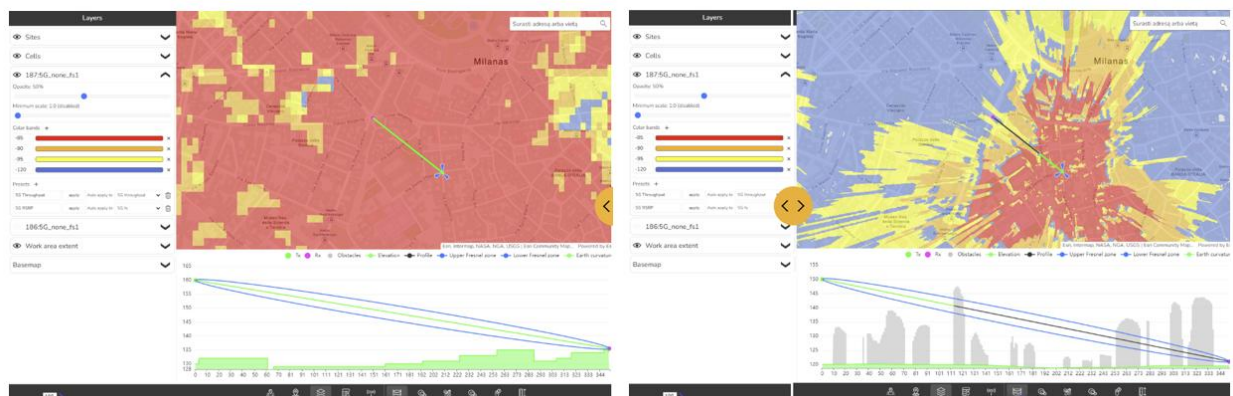


The CE tools make use of three distinct GIS data layers to obtain high precision modelling of radio wave propagation losses:

- **Digital Terrain Model (DTM)**, also known as Digital Elevation Model (DEM), which describes Earth surface, i.e., path terrain profile in terms of ground elevation above uniform sea level.
- **Obstacles layer**, delineating buildings and other such objects above Earth surface that may be considered to be principal impediments for radio wave propagation.
- **Clutter layer**, delineating natural occurring or human cultivated ground cover that may be partially penetrable by radio waves, such as natural vegetation (e.g., forests, trees, bushes) or various crops, gardens, parks, etc.

The image above illustrates how Cellular Expert uses different resolutions of topographical data to significantly improve coverage prediction accuracy.

- Left image: Coverage calculated using 25 m resolution ASTER DEM data, showing a general view of signal distribution but with limited detail, especially in dense urban areas.
- Right image: Coverage calculated using 1 m resolution data, revealing a much more precise propagation pattern, including building-level shadowing and accurate street-by-street coverage.

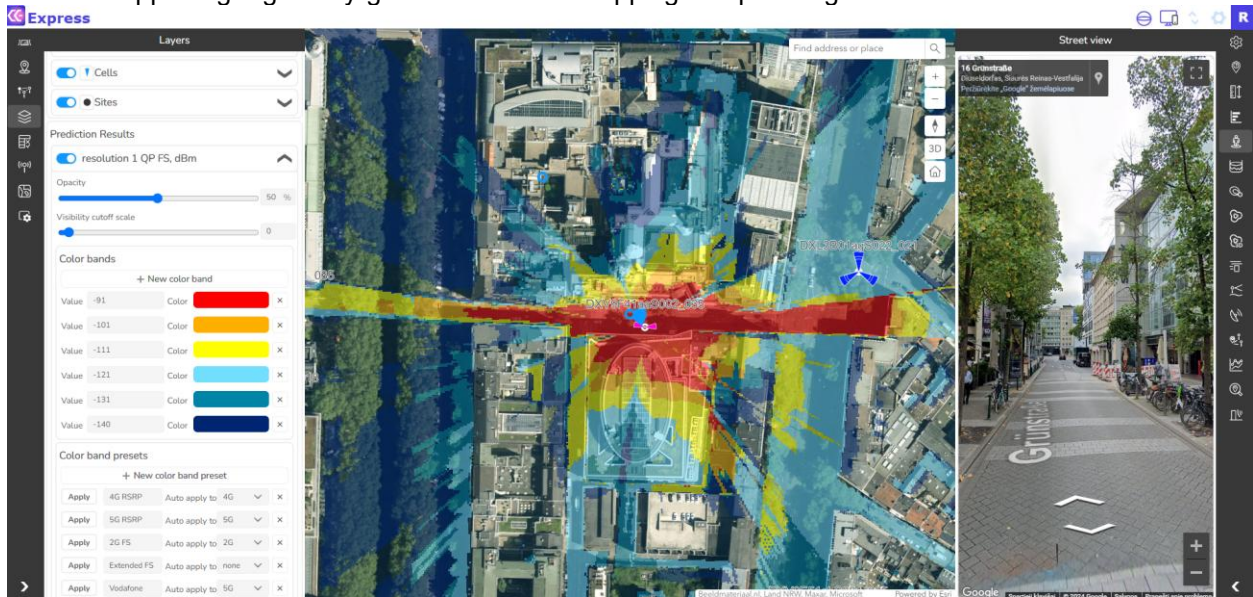


More information: <https://blog.maxar.com/earth-intelligence/2022/benefits-of-using-maxars-precision3d-telco-suite-for-5g>

Cellular Expert can easily integrate and process 1 m or even sub-meter topographical data, providing highly detailed RF calculations. This level of precision is essential for:

- Modeling 2G/3G/4G/5G, small cells and mmWave networks.
- Identifying exact coverage gaps at the building and street level.

- Supporting regulatory-grade broadband mapping and planning.



By using high-resolution terrain and clutter data, Cellular Expert ensures that its calculations match real-world conditions as closely as possible — resulting in better network design decisions and more reliable broadband planning outcomes.

2. Geographic data requirements

The supported geographical data types:

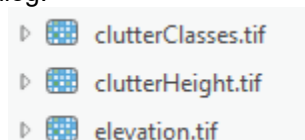
Only GeoTIFF is supported. Topographical data must have specific names:

- The Digital terrain model must be named *elevation.tif*
- The land use (or clutter) grid must be named *clutterClasses.tif*
- The clutter heights (typically building, vegetation height) grid must be named *clutterHeight.tif*

Mandatory geographical data:

- Digital Terrain Model (DTM) grid

All geodata must be located in one catalog.



2.1 Digital Terrain Model (DTM) Grid (Mandatory)

The Digital Terrain Model (DTM), also known as Digital Elevation Model (DEM), represents the Earth's ground level above sea level. Each raster pixel has its height value.

A sample DTM raster is presented below. Each pixel represents height value above the sea level. In reality, within a one-pixel area, the height is not the same everywhere. Thus, the pixel's height value is the height

in its center or the maximum. The smaller the pixels, the more accurate is the grid - but also more data to calculate.



2.1.1 Projection

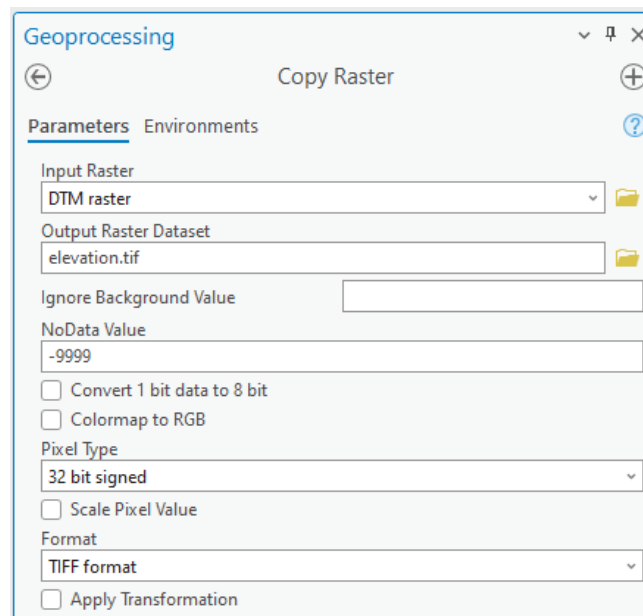
The raster must use a Projected Coordinate System. To check the coordinate system of your raster, use the Properties function in ArcGIS Pro. Add the raster to your project, right-click on it, and select Properties. Then, go to the Source tab > Spatial Reference and check the Coordinate System type parameter to confirm it is in a Projected Coordinate System.

2.1.2 Correct No Data value and raster name

After setting the correct projection, assign the NoData: -9999 attribute and specify the appropriate name for the DTM raster.

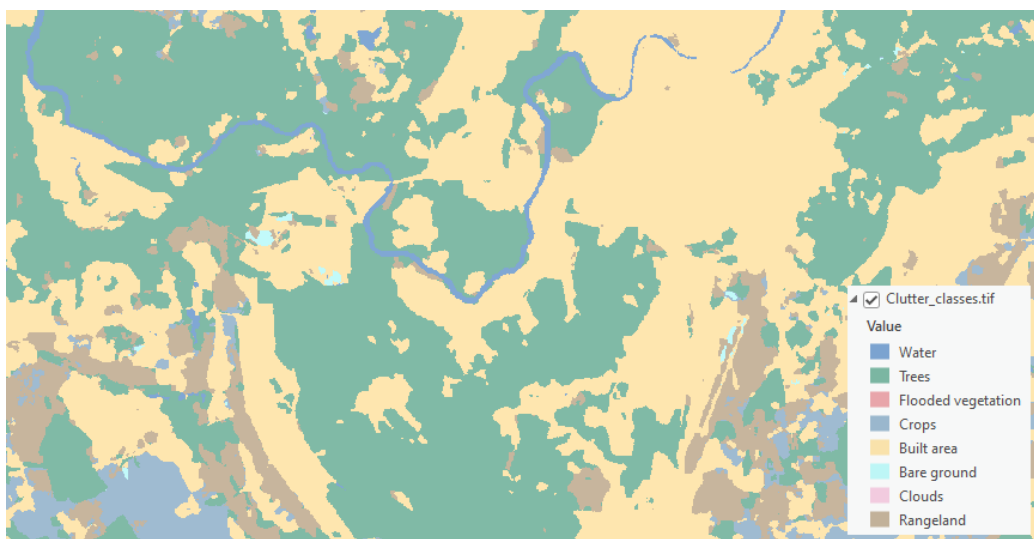
2.1.3 Pixel type

16-bit signed, or 32-bit signed or 32-bit float



2.2 Clutter classes grid (optional)

Land use or clutter refers to the classification of the earth's surface into categories such as urban, suburban, rural, forest, water, and open land, each of which affects radio propagation differently. Clutter data is crucial because it determines how signals are absorbed, reflected, or diffracted by the environment, directly influencing coverage, interference, and quality of service. The naming and classification of land use types may vary. An example is the Sentinel-2 Land Cover dataset from the Living Atlas: [Living Atlas Sentinel-2 Land Cover](#)



2.2.1 Projection

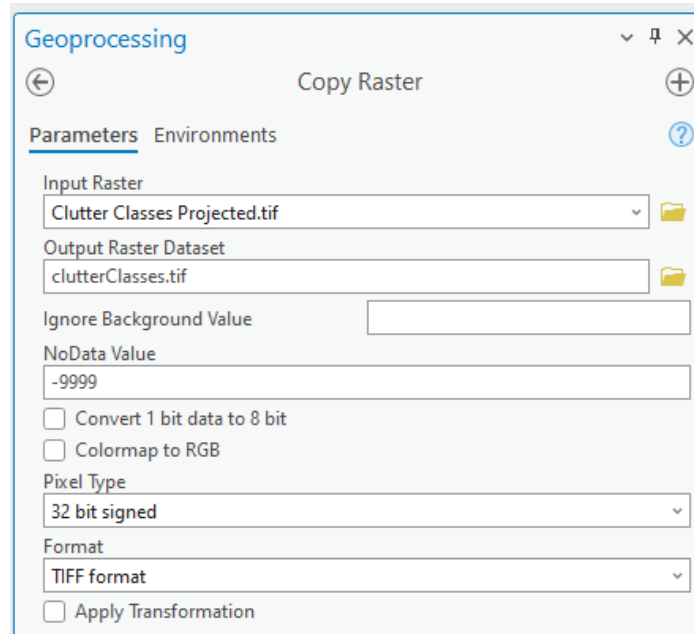
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2.2.3 Pixel type

16-bit signed, or 32-bit signed or 32-bit float

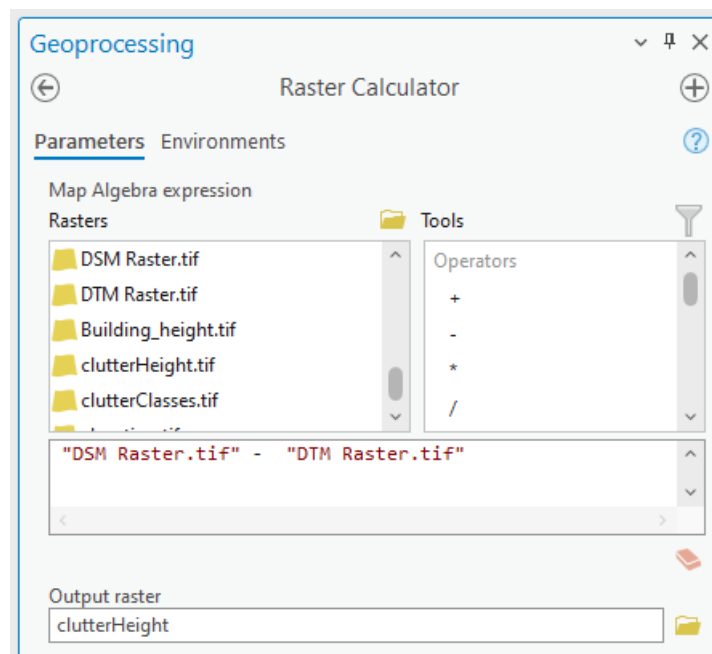


2.3 Clutter heights (optional)

Clutter heights represent the vertical dimension of obstacles such as buildings, trees, and other surface features **above** the digital terrain model (DTM). While the DTM provides the bare-earth elevation, clutter heights add the true 3D profile of the environment by capturing how high objects rise above the ground. The clutter heights raster requires the accompanying clutterClasses.tif raster and cannot be used independently.



A clutter height raster can be derived from a Digital Surface Model (DSM) raster and a Digital Terrain Model (DTM) raster using the ArcGIS Raster Calculator tool.



The calculation output will be the difference between the DSM and DTM grids, representing the clutter heights.

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